

The (Un)making of Data Subjects in Digital Credit Repair Practice

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Abstract

This research project investigates the techno-politics of digital credit repair practices, defined as the visions and practices that digital credit companies and borrowers draw on to sustain or contest their credit relationships. On one hand, digital credit companies leverage AI to ‘see’ and amplify certain patterns in consumer data, transforming borrowers into data subjects within regimes of data capture and prediction. On the other hand, borrowers draw on culturally situated practices to achieve ‘good’ credit scores, though these practices may be labeled as attempts to game the system. Using Kenya as a field site, this project aims to examine the tensions emerging in digital credit relationships, paying close attention to the artful work involved in translating and repairing relations and exchanges between digital credit companies and borrowers.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *HCI theory, concepts and models*.

Keywords

credit repair, credit scoring, digital loans, fintech, infrastructure, mobile money

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1 Introduction

In Nairobi, Kenya, new experimental models of credit scoring—often developed using emerging technologies such as artificial intelligence (AI) and big data—are being used to create infrastructure for digital credit, ostensibly de-risking the unbanked and majority poor [38, 44]. These models are additionally serving as laboratories for local and global circulation of digital credit [56]. Since Kenya and several countries in Africa do not have standardized consumer credit scoring systems like the United States’s FICO [27], different financial technology (fintech) companies are drawing on troves

of transactional and consumer data to develop alternative credit scoring technologies, expanding access to consumer credit [44].

Yet there are growing ethical and practical concerns around the design and deployment of credit scoring systems [38, 40]. On one hand, fintechs in Kenya have been accused of data privacy abuses, ballooning interest rates, and even using public shaming tactics in loan recoveries [31, 44]. On the other hand, there has been recent attention towards the counter-tactics and cultural practices that borrowers are adopting to navigate their credit obligations to digital lenders—including changing SIM (identity) cards or refinancing digital loans from borrowing elsewhere [38]. These concerns have been a source of tension between fintechs and borrowers. To examine these tensions, my research draws on digital credit repair, which I define as the visions and practices that fintech companies and their customers mobilize to sustain or contest their credit relationships. What can the shifting registers of knowledge and values between fintechs and borrowers tell us about credit scoring and the future of digital lending in Kenya and beyond?

Using qualitative and design research methods, my research aims to examine digital credit repair practices in Kenya by critically engaging with the use of AI in fintech and the recent imbrication of these technologies into microcredit worlds, popularized under the banner of “financial inclusion.” As such, my project seeks to show how machines could ‘see’ and transform debtors into data subjects, assessing their risk profiles and predicting their credit-worthiness. At the same time, my project also aims to highlight the local and situated user credit cultures and practices that emerge in response, and the artful work that goes into translating and repairing exchanges between fintechs and their users.

2 Context and motivation

The promise of AI and recent transformations in mobile telecommunications have captured popular interest and imagination in computing and development circles as much as microfinance did two decades ago [37]. Founded and evangelized by Nobel Laureate Muhammad Yunus in Bangladesh in the 1970s, microfinance was marketed as a key solution to the design challenge of poverty. While the microfinance model has been replicated in multiple development initiatives in the global majority world for decades [21], it has not lifted 100 million people out of poverty as hoped. This has been attributed to the model’s financialization of the poor, which has exacerbated indebtedness and despair, even causing death [12, 46]. To address pressing and urgent concerns about microfinance’s challenges, powerful actors in the private sector such as telecommunications and banking, a range of multilateral development entities including the World Bank and Gates Foundation, academic institutions, and a coalition of governments in the developing world have

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drawn on recent social, economic, and technological transformations to recast the failures of microfinance as technical challenges that can be addressed through big data and AI-enabled credit scoring to de-risk the unbanked in the global majority world.

While my research project builds on debates that have characterized technical interventions to problems of global development, foregrounding discussions within and beyond HCI [8, 9, 11, 18, 20, 51], it adopts a broader view of the infrastructures and politics of digital credit and their relationship to global development. By showing how data and AI infrastructures are enrolled in the attempts to address historically complex challenges associated with microfinance and global development, my project examines the definitions of good practice and behavior in digital credit worlds, locating concerns within credit scoring systems—the technologies of classification and prediction often developed from murky data sources—and how the attempts by users to manage “good” credit scores might be misunderstood as adversarial in technical practice [58]. Whereas AI-generated credit scoring systems might certainly reward or penalize certain behaviors in digital credit worlds,¹ however, the true range of possibilities for good or for bad is poorly understood: there has been little cross-disciplinary dialog and public engagement. The subject of my work is, thus, how citizens are transformed as data subjects in these sociotechnical AI infrastructures—instead of how citizens are excluded from these systems—and the tensions emerging from these transformations, including how they are resolved or not.

3 Key related work

My project contributes to recent discussions on AI and global development [4], marked by the promises and roles for AI in social change and economic transformation [10]. By offering historically grounded and contemporary accounts of AI in material and cultural exchange and production, my project unlocks new ways to account for the broad and downstream implications of technology across heterogeneous contexts and cultures. To do so, I build on a theoretically rich empirical program across three disparate but interlinked themes: critical data and AI studies, repair and care, and infrastructure and global development. Drawing on foundational and emerging lines of inquiry in HCI, STS, anthropology, philosophy, and behavioral economics, my project emphasizes the political and normative concerns in digital credit worlds, and how fintechs and borrowers sustain or contest their credit relationships.

3.1 Critical data and AI studies

Human-centered studies of AI, machine learning (ML), and data science have expanded dramatically in HCI and CSCW in the recent past. In dialog with allied fields including fairness, accountability, and transparency (FAccT) and the emerging program of critical AI, researchers have scrutinized the downstream impacts of AI/ML, raising real questions about processes of data collection and annotation [48], the values in model design and production [6], and implications of AI in deployment [2], especially in high-stakes prediction and decision domains including credit scoring [22, 45, 52],

¹If microfinance faced challenges due to poor enforcement of credit contracts and ‘irrational’ borrowers who used microcredit for ‘unproductive’ purposes, then these new AI-driven arrangements promise to discipline consumption at a distance while also addressing grand societal challenges.

healthcare [36, 41], criminal justice [3, 28], and social welfare [50]. Although most of this work is based on Euro-American contexts, a small and growing body of scholarship has foregrounded global majority contexts (especially Africa and South Asia), generating useful insights on the potential social and economic benefits of AI [10] while also exposing the hidden labor and affective work that makes AI actually work [4, 47, 48], as well as calling out other forms of marginalization and exclusion in global AI development and regulation [42]. My approach follows what Douglas-Jones et al. [13] recognize as data’s (and indeed AI’s) transformative potential, “its capacity to produce new social compositions while indexing the old.” However, it also raises questions about values, logics, and worldviews embedded in the design and application of AI and data-driven systems. Critical scholars of technology have already shown how marginalized and low-income communities have been the first to experience technology, where it is experimented on them and perfected for the rest of society [5, 16, 33]. *Cui bono?* At whose expense?

3.2 Repair and care

HCI and CSCW researchers have in the past decade generated a large body of empirical and theoretical work on repair, some of it animated by what Jackson [24] has called “broken world thinking,” taking breakdown and decay rather than “novelty, growth, and progress” as the starting point in “(re)thinking technologies, infrastructures, and their relations to the social and material worlds.” Matters of repair are commingled with the logic of care: “everything that we do to maintain, continue and repair ‘our world’ so that we can live in it as well as possible” [55]. My project uses the lens of repair and care to better understand credit scoring as an institutional practice, socially and practically accomplished, and the responses it elicits from borrowers and customers. HCI scholars have called out the institutional and affective work that makes these infrastructures functional, the significant on-ground labor constituted in the workarounds that get things back on track [54]. Some of these workarounds include the invisible professional labor that makes newly deployed machine learning systems function across different institutional and organizational contexts [15], and the imperatives balanced in navigating “trustworthiness” of data analytics in the face of uncertainty [39, 40]. At the same time, my approach to customer responses to credit scoring is centered around symbolic and material practices that users draw on to renegotiate their credit relationships with digital lenders (See also related practices mobilized through ‘hacking’ and ‘appropriation’ [32]). Akrich’s [1] case studies in West Africa show how tiny practices of “de-description” contributed to the collapse of carefully designed networks of technology. In digital credit worlds, fintech companies tend to portray these user practices as attempts to “game the system” [58], reclaiming them back into the system through processes of enclosure or even banishing them altogether for violating terms of service. Ultimately, the entanglement between fintech companies actors and their customers should not be seen in one direction “but as a recursive loop between the calculations of the algorithm and the ‘calculations’ of people” [19].

3.3 Infrastructure and global development

Star and Ruhleder [53] in a seminal CSCW paper defined infrastructure as relational, and not just a thing, emphasizing its historical and political contingencies. Instead of asking what is infrastructure, Star and Ruhleder behoove us to ask when is infrastructure, generating interest about when a thing becomes a valid relational practice. Since Star's pathbreaking work, a large body of work in HCI as well as in STS and media studies has extensively examined infrastructures in their recursions and dispersions across contexts and geographies, shedding light on the cultural and technical practices through which work and social action is accomplished [20, 49, 59]. Beyond animating social action, infrastructures engender visions and desires of alternate futures [26]. My project takes the case of M-Pesa,² an electronic money exchange platform first successfully deployed in Kenya in 2007 to highlight how the platform has become the "rails" on which a broader set of financial services can ride [29, 30] and how it has trailblazed a new form of imagination in global development [8, 32]. Yet, despite its successes, M-Pesa—and the digital credit products built on top of it—might mirror problematic dynamics between north-south relations.

Using a postcolonial computing lens, Irani et al. [23] seek to bring to bear the uneven technical and economic relations dominant in digital development, critiquing assumptions of the global majority as "powerless and unable to 'develop' without intervention." Digital development interventions, which HCI4D and ICTD engage with and contribute to, have often de-politicized thorny social and economic problems like poverty by proposing minute "[technical], behavioral, educational, and market-based solutions" [23]. My project is alive to these dynamics and contradictions. Whereas it broadly attempts to recover dimensions of the cultural often sidelined in technical ML/AI research [17], it recognizes "the depths of inter-dependence of [infrastructures]...and the real work of politics and knowledge production" [7]. On this account, my project leverages the tactic of "infrastructural inversion" to surface what is usually relegated to the invisible [7]; deconstructing the design choices of technical, social, and historical import and the institutional arrangements that undergird them, and everyday practices that sustain or challenge deployments of these technologies.

4 Specific research questions

The primary research questions that animate this project are the following: 1) How and to what extent are (historical) visions and practices of credit access reflected in contemporary credit relationships between creditors and borrowers? 2) What instantiates tensions in these credit relationships, and how are such tensions repaired by creditors and borrowers? 3) How are credit relationships and credit repair practices changing with new actors and technologies in the digital age? 4) How are borrowers drawing on shifting configurations of digital lending to imagine alternative futures?

²Although M-Pesa was initially conceptualized as a microfinance project, its redesign and development as a mobile money platform has been the center of attention in HCI and design and policy circles as the proper way to do digital development.

5 Research methods and rationale

I will employ archival, ethnographic, and speculative design methods in my research. My approach to the study of digital credit repair doubly builds on a materialist understanding of infrastructural dynamics and the moral uncertainties of digital credit markets. In my work, I will show how digital lenders leverage data science, product development, and other sociotechnical practices to develop credit scoring systems. At the same time, I will detail how borrowers are responding, thus renegotiating their credit relations and obligations to digital lenders. Against this background, the project seeks to address four key research questions (see section 4), using different but related methodological approaches—archival, ethnographic, and design research.

I will use archival methods to study how lenders and debtors in Kenya have historically drawn on different registers of knowledge and values to instantiate certain expectations of propriety or mutual obligation across exchanges and transactions. Using rich secondary sources including policy documents, case law, newspaper articles, web blogs, and online videos, I will show connections between historical and contemporary credit practices, and the tensions emerging. For example, the use of shaming tactics by digital lenders in loan recoveries in Kenya [31] is similar to microfinance's enduring instrumentalization of gendered codes of shame in enforcing credit repayments [25]. I will also show the range of historically grounded counter-tactics that borrowers have adopted to explicitly or tacitly renegotiate their obligations to lenders [34, 38]. The tensions emerging in these digital credit relationships have for example prompted regulatory responses from the Kenyan government including amendments to the recently introduced digital lending licensing framework and from Google, which has imposed restrictions to regulate the deluge of digital credit apps in the Play Store [35]. To extend this archival research, I will draw on oral histories of policy and regulatory practitioners, accounting for the political and economic grounds through which state and non-state actors have made credit a necessary and aspirational part of everyday life, and how ordinary people have used (or failed to use) such credit within their systems of meaning-making and financial citizenship.

Using ethnographic methods, I will study digital credit repair practices between a digital lender and borrowers in Nairobi, describing encounters between corporate AI-enabled digital lending and historically layered user credit cultures and practices. By getting "inside" an established digital lender as a UX intern or researcher, I will conduct "real-time" observations of the organizational and affective work of data science and product marketing teams in developing proprietary credit scoring systems. Likewise, I will engage consumers of fintech products and services to learn about their responses. My project will study the digital credit relationships between the digital lender and users, borrowing from growing ethnographic work in HCI and adjacent fields that have separately examined, for instance, the construction of credit scoring systems by data analytics companies [40] and how users cope with or resist such algorithmic systems [2, 43, 58].

I will also use speculative design methods, defined as a reflexive practice that pursues debate about dominant worldviews or cultural values in design, opening up the spaces of possibility to design

without moorings of market demands and social and moral conventions [14]. In my project, I will host two co-design workshops to extend speculative design to the realm of digital credit, co-designing with participants, for instance, what an alternative values-centered credit scoring system could be. The first workshop will build an understanding of the dynamics and contradictions of credit scoring and the second will generate future alternatives to digital credit products. The goal is to shift treating AI-enabled digital lending as a fixed system and instead treat it as a dynamic infrastructure, open to debate and change. Examples of relevant design activities include making low-fidelity prototypes, technology and cultural probes, and social network diagrams. Throughout the design process, I will also document the outcomes of the design activities by taking photos, videos, and notes. After the design workshops, I will compile the outputs into a design workbook that will be shared with workshop participants [57].

Collectively, these methods will be leveraged to develop a deeper understanding of digital credit repair practices and chart paths to constructing alternative futures in digital credit worlds.

6 Research progress and contributions to date

Broadly, I have studied the design and governance of sociotechnical infrastructures, as well as the benefits and costs these infrastructures produce. My PhD work, published in HCI venues, intersects computing and development in at least two ways. First, I have reexamined the notion of appropriation in global HCI [32], which is broadly defined as the “unexpected adaptations or modifications of artifacts performed by users in concert with others and with things.” By investigating the story of airtime exchange, early practices where mobile phone users in Africa creatively misused telecommunications networks by trading airtime (or phone credit) vouchers for cash, I show that the enclosure of these informal practices into standardized products like mobile money was only one story of appropriation. Before informal airtime exchange practices, telecommunications operators appropriated the ‘GSM’ (a global telecommunications standard) to global majority world markets kickstarting a series of appropriations. Accordingly, I showed that appropriation is not just about the initial uses of technology where “end-users” make “technology theirs” but rather unfolds dialectically, imbricating a range of actors including ordinary users, and state and non-state actors.

Second, I have developed new perspectives on “moral orders of repair,” which I define as the norms, rules, and expectations that structure everyday practices, exchanges, and collaborations in repair worlds and other spheres of practice (under review). Developed from a recent extended ethnographic field visit in Kampala, Uganda I investigated the moral orders of repair in the mobile phone and personal computing repair worlds in Kampala, Uganda showing how moral orders are co-constituted within and through the practical and ethical work of repair, maintenance, and collaboration.

7 Expected next steps

The expected next step for my dissertation research is to complete the necessary fieldwork for my project. The fieldwork for my dissertation will take place in two phases for at least six months. In the first phase, across three months, I will be embedded as a UX intern

or researcher in a digital lending company in Kenya where I will study the making of credit scoring systems as a sociotechnical practice. In the remainder of the time, I will engage with participants in Nairobi to build shared knowledge of AI-enabled credit scoring and chart paths toward more equitable and inclusive alternatives.

8 Dissertation status and long-term goals

My dissertation is in progress, and I expect to complete it in 2027. As mentioned, I will embark on my fieldwork in Kenya where I aim to conduct a series of participant observations, semi-structured interviews, and design research. Post-fieldwork, I expect to publish in conferences such as CHI, CSCW, and FAccT, examining empirical and theoretical aspects of digital credit repair practice in Kenya and beyond. During this time, I also plan to enter the job market where I will seek research and teaching positions in academic institutions or research-oriented roles in industry or civil society. Ultimately, my long-term goal is to build a community-centered research program, aiming to understand the benefits and costs of data and AI infrastructures in high-stakes domains as well as emphasizing the role of design for social change.

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